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Negative Binomial Kumaraswamy-G Cure Rate Regression Model

by Amanda D'Andrea [1,2,*], Ricardo Rocha [3], Vera Tomazella [1] and Francisco Louzada [2]

[1] Department of Statistics, Federal University of São Carlos, São Carlos, SP 13565-905, Brazil

[2] Institute of Mathematical Science and Computing, University of São Paulo, São Carlos, SP 13565-905, Brazil

[3] Department of Statistics, Institute of Mathematics and Statistics, Federal University of Bahia, Salvador, BA 40170-115, Brazil

[*] Author to whom correspondence should be addressed.

J. Risk Financ. Manag. **2018**, *11*(1), 6; https://doi.org/10.3390/jrfm11010006

Submission received: 8 December 2017 / Revised: 15 January 2018 / Accepted: 16 January 2018 / Published: 19 January 2018

(This article belongs to the Special Issue Extreme Values and Financial Risk)

Abstract

In survival analysis, the presence of elements not susceptible to the event of interest is very common. These elements lead to what is called a fraction cure, cure rate, or even long-term survivors. In this paper, we propose a unified approach using the negative binomial distribution for modeling cure rates under the Kumaraswamy family of distributions. The estimation is made by maximum likelihood. We checked the maximum likelihood asymptotic properties through some simulation setups. Furthermore, we propose an estimation strategy based on the Negative Binomial Kumaraswamy-G generalized linear model. Finally, we illustrate the distributions proposed using a real data set related to health risk.

Keywords: long-term survivors; Kumaraswamy family; survival analysis; negative binomial distribution; generalized linear model

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Comprovante do artigo JAS 2018 - página/capa bibliográfica

Categoria: Artigos

Regra: 2.9 - 15 pontos por publicação

Pontos lançados: 15.00

Journal of Applied Statistics
Volume 46, 2019 - Issue 3

Articles

Defective models induced by gamma frailty term for survival data with cured fraction

Juliana Scudilio, Vinicius F. Calsavara, Ricardo Rocha, Francisco Louzada, Vera Tomazella & Agatha S. Rodrigues

Pages 484-507 | Received 03 Aug 2017, Accepted 28 Jun 2018, Published online: 18 Jul 2018

Cite this article https://doi.org/10.1080/02664763.2018.1498464

ABSTRACT

In this paper, we propose a defective model induced by a frailty term for modeling the proportion of cured. Unlike most of the cure rate models, defective models have advantage of modeling the cure rate without adding any extra parameter in model. The introduction of an unobserved heterogeneity among individuals has bring advantages for the estimated model. The influence of unobserved covariates is incorporated using a proportional hazard model. The frailty term assumed to follow a gamma distribution is introduced on the hazard rate to control the unobservable heterogeneity of the patients. We assume that the baseline distribution follows a Gompertz and inverse Gaussian defective distributions. Thus we propose and discuss two defective distributions: the defective gamma-Gompertz and gamma-inverse Gaussian regression models. Simulation studies are performed to verify the asymptotic properties of the maximum likelihood

Comprovante do artigo Biometrical Journal 2019

Categoria: Artigos

Regra: 2.9 - 15 pontos por publicação

Pontos lançados: 15.00

Item 2.9

Received: 12 March 2018 | Revised: 17 October 2018 | Accepted: 22 February 2019

DOI: 10.1002/bimj.201800056

Biometrical Journal

RESEARCH PAPER

Defective regression models for cure rate modeling with interval-censored data

Vinicius F. Calsavara[1] | Agatha S. Rodrigues[2,3] | Ricardo Rocha[4] | Vera Tomazella[5] | Francisco Louzada[6]

[1]Department of Epidemiology and Statistics, A.C. Camargo Cancer Center, São Paulo, SP, Brazil

[2]Institute of Mathematics and Statistics, University of São Paulo, São Paulo, SP, Brazil

[3]Department of Obstetrics and Gynecology, São Paulo University Medical School, São Paulo, SP, Brazil

[4]Department of Statistics, Federal University of Bahia, Salvador, BA, Brazil

[5]Department of Statistics, Federal University of São Carlos, São Carlos, SP, Brazil

[6]Institute of Mathematical Science and Computing, University of São Paulo, São Carlos, SP, Brazil

Correspondence
Vinicius F. Calsavara, Department of Epidemiology and Statistics, A.C. Camargo Cancer Center, São Paulo, SP, Brazil.
Email: vinicius.calsavara@accamargo.org.br

Abstract

Regression models in survival analysis are most commonly applied for right-censored survival data. In some situations, the time to the event is not exactly observed, although it is known that the event occurred between two observed times. In practice, the moment of observation is frequently taken as the event occurrence time, and the interval-censored mechanism is ignored. We present a cure rate defective model for interval-censored event-time data. The defective distribution is characterized by a density function whose integration assumes a value less than one when the parameter domain differs from the usual domain. We use the Gompertz and inverse Gaussian defective distributions to model data containing cured elements and estimate parameters using the maximum likelihood estimation procedure. We evaluate the performance of the proposed models using Monte Carlo simulation studies. Practical relevance of the models is illustrated by applying datasets on ovarian cancer recurrence and oral lesions in children after liver transplantation, both of which were derived from studies performed at A.C. Camargo Cancer Center in São Paulo, Brazil.

KEYWORDS

defective distribution, Gompertz distribution, inverse Gaussian distribution, interval-censored data, long-term survivor

1 | INTRODUCTION

In survival data, it is often of interest to model the time T until an event (e.g., cancer recurrence) occurs; but in some situations, time T may only be known to have occurred within an interval $(U, V]$, where $U < T \leq V$. For instance, a liver transplant patient may develop an oral lesion that is detected during a follow-up visit after surgery. In this situations, patients submitted to liver transplant are assessed at clinic visits by a dentist in order to detect oral lesion. The time to the event (oral lesion) is not observed, but the event is known to have happened between time U and time V.

In practice, researchers typically perform traditional survival analysis methods using the date when the event is diagnosed (diagnostic exam) as the moment of event occurrence. However, if an event occurs before time V but after time U, ignoring the interval-censored mechanism in survival analysis can lead to overestimates in survival curves and erroneous conclusions (Rodrigues, Calsavara, Silva, Alves, & Vivas, 2018).

Researchers have proposed some approaches for handling such interval-censored data. Peto (1973) provided an estimation method for the cumulative distribution function, which is similar to the life-table technique, as well as the self-consistency algorithm for the estimation of the survival function proposed by Turnbull (1976). Semiparametric approaches based on the

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Categoria: Artigos

Regra: 2.9 - 15 pontos por publicação

Pontos lançados: 15.00

JOURNAL OF APPLIED STATISTICS
https://doi.org/10.1080/02664763.2019.1597029

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Zero-adjusted defective regression models for modeling lifetime data

Vinicius F. Calsavara [a], Agatha S. Rodrigues [b,c], Ricardo Rocha [d], Francisco Louzada [e], Vera Tomazella [f], Ana C. R. L. A. Souza [c], Rafaela A. Costa [c] and Rossana P. V. Francisco [c]

[a]Department of Epidemiology and Statistics, A.C.Camargo Cancer Center, São Paulo, SP, Brazil; [b]Institute of Mathematics and Statistics, University of São Paulo, São Paulo, SP, Brazil; [c]Department of Obstetrics and Gynecology, São Paulo University Medical School, São Paulo, SP, Brazil; [d]Department of Statistics, Federal University of Bahia, Salvador, BA, Brazil; [e]Institute of Mathematical Science and Computing, University of São Paulo, São Carlos, SP, Brazil; [f]Department of Statistics, Federal University of São Carlos, São Carlos, SP, Brazil

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ABSTRACT

In this paper, we introduce a defective regression model for survival data modeling with a proportion of early failures or zero-adjusted. Our approach enables us to accommodate three types of units, that is, patients with 'zero' survival times (early failures) and those who are susceptible or not susceptible to the event of interest. Defective distributions are obtained from standard distributions by changing the domain of the parameters of the latter in such a way that their survival functions are limited to $p \in (0, 1)$. We consider the Gompertz and inverse Gaussian defective distributions, which allow modeling of data containing a cure fraction. Parameter estimation is performed by maximum likelihood estimation, and Monte Carlo simulation studies are conducted to evaluate the performance of the proposed models. We illustrate the practical relevance of the proposed models on two real data sets. The first is from a study of occlusion of endoscopic stenting in patients with pancreatic cancer performed at A.C.Camargo Cancer Center, and the other is from a study on insulin use in pregnant women diagnosed with gestational diabetes performed at São Paulo University Medical School. Both studies were performed in São Paulo, Brazil.

ARTICLE HISTORY
Received 24 October 2018
Accepted 14 March 2019

KEYWORDS
Defective distribution; Gompertz distribution; inverse Gaussian distribution; long-term survivor; zero-adjusted

1. Introduction

In medical studies, it is common that some units under study are not susceptible to the event of interest, called immune or cured elements. A class of models, referred to as cure rate models, considers these situations and has been studied by several authors in the recent years. The most popular model is the standard mixture model, which was first proposed by Boag [5] and Berkson and Gage [4]. In this model, the population survival function is $S(t) = p_1 + (1 - p_1)S_0(t)$, such that $p_1 \in (0, 1)$ is referred to as the cured fraction, and $S_0(t)$ is a proper survival function for uncured patients. Standard mixture model is widely

CONTACT Vinicius F. Calsavara vinicius.calsavara@accamargo.org.br